

FORMATTING INSTRUCTIONS FOR ICLR 2027

CONFERENCE SUBMISSIONS

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ABSTRACT

1 INTRODUCTION

2 RELATED WORKS

3 PRELIMINARIES

3.1 MULTI-AGENT COORDINATION AS BELIEF-CONDITIONED SEQUENTIAL DECISION MAKING

In sequential multi-agent decision making, the consequence of an ego action depends not only on the physical state but also on how other agents respond. This dependence is particularly important when a partner’s goals, incentives, or other behaviorally relevant information are not directly observable. We therefore adopt the interactive POMDP (I-POMDP) perspective, which incorporates models of other agents into sequential decision making and maps beliefs over interactive states to actions (Gmytrasiewicz & Doshi, 2005).

Following the canonical definition of Gmytrasiewicz and Doshi, an I-POMDP for ego agent i is

$$\text{I-POMDP}_i = \langle IS_i, A, T_i, \Omega_i, O_i, R_i \rangle, \quad IS_i = S \times \prod_{j \neq i} M_j, \quad A = \prod_{j=1}^N A_j, \quad (1)$$

where S is the physical state space, M_j is the set of possible models of partner j , and A is the joint action space. In the original two-agent presentation, $IS_i = S \times M_j$ and $A = A_i \times A_j$; the product form above is its direct multi-agent generalization. Each model $m_j \in M_j$ is described by $m_j = \langle h_j, f_j, O_j \rangle$, where h_j is a possible observation history, $f_j : H_j \rightarrow \Delta(A_j)$ maps that history to a distribution over j ’s actions, and O_j specifies the input supplied to j by the environment. Models may be subintentional, M_j^S , or intentional, M_j^I , so that $M_j = M_j^I \cup M_j^S$. An intentional model is a type $\theta_j = \langle b_j, \hat{\theta}_j \rangle$, where b_j is j ’s belief over the physical state and other-agent models and $\hat{\theta}_j$ is its frame, including its action, observation, transition, reward, and optimality specifications. Thus, the recursive belief structure is part of the standard model space rather than an additional assumption introduced here (Gmytrasiewicz & Doshi, 2005).

The remaining components are the transition, observation, and reward models,

$$T_i : S \times A \times S \rightarrow [0, 1], \quad O_i : S \times A \times \Omega_i \rightarrow [0, 1], \quad R_i : IS_i \times A \rightarrow \mathbb{R}. \quad (2)$$

The original formulation imposes model non-manipulability (MNM): an agent’s actions do not directly modify another agent’s model, and model non-observability (MNO): an agent cannot directly observe another agent’s model. Consequently, an agent maintains a belief $b_i^t \in \Delta(IS_i)$ over interactive states and updates it from its own action and observation. Using the belief-state estimation operator from the original formulation, this update is

$$b_{t+1}^i = \text{SE}_{\theta_i}(b_t^i, a_t^i, o_{t+1}^i) \equiv \tau_i(b_t^i, a_t^i, o_{t+1}^i). \quad (3)$$

At this level, SE_{θ_i} is not a filter over a static partner label: for an intentional model, the original update marginalizes over partner actions and observations, the physical transition, and the partner’s own belief update. This recursive dependence is precisely what distinguishes an interactive belief from a flat POMDP state belief.

For an intentional partner model, the expected immediate reward in the value recursion is averaged over the partner actions predicted by its model. In the two-agent definition, this quantity is

$$\text{ER}_i(x, a_i) = \sum_{a_j \in A_j} R_i(x, (a_i, a_j)) \Pr(a_j \mid m_j), \quad (4)$$

and is replaced by an expectation over the joint action a_{-i} for multiple partners. The canonical I-POMDP solution maps the agent’s type to an action through

$$U_i(\theta_i) = \max_{a_i \in A_i} \left[\sum_{x \in IS_i} b_i(x) \text{ER}_i(x, a_i) + \gamma \sum_{o_i \in \Omega_i} \Pr(o_i \mid a_i, b_i) U_i(\langle \text{SE}_{\theta_i}(b_i, a_i, o_i), \hat{\theta}_i \rangle) \right]. \quad (5)$$

Here $\Pr(o_i \mid a_i, b_i)$ is induced by T_i , O_i , and the models of the other agents. The belief update and value recursion above are the standard I-POMDP mechanisms: the former updates a distribution over interactive states, while the latter selects an ego action after integrating over the predicted behavior of other agents. For the finite-horizon environment used here, we instantiate the same dependency with undiscounted terminal utility and no separately shaped intermediate reward.

The two terms below are therefore not additional state variables or a new coordination formalism. They are functional views of the standard interactive belief and solution mapping. For the sequential notation used below, s_t denotes the public interaction state, a_t^i an ego action, r_{t+1}^j a response emitted by partner j when an interaction elicits one, o_t^i the observation available to ego i , and e_{t+1}^i the evidence extracted from the next observed interaction.

A *partner belief* is the ego’s current belief about the partner components of the interactive state: equivalently, the uncertainty over the partners’ models or types that is relevant to predicting their future behavior. In the original I-POMDP, this object is represented within b_t^i and can include partners’ private preferences, capabilities, observations, and beliefs. In an implementation, its representation may be a finite support, a structured state, or an implicit internal representation; it need not be exposed as an explicit probability vector or required as a model output. This use of a behavioral model revised from observed input–output behavior is also standard in opponent modeling (Carmel & Markovitch, 1996).

Partner-conditioned planning is the complementary use of that belief in the action-selection part of the I-POMDP solution. Given b_t^i , the ego uses the models in the support of its belief to predict partner responses, e.g., $r_{t+1}^j \sim \hat{\pi}_j^i(\cdot \mid s_t, a_t^i, b_t^i)$, evaluates the resulting downstream utility, and selects its own action. The candidate action may determine which partner to engage and what interaction to propose; when the ego is responding, the response itself is its action. This functional view is consistent with prior work that couples inferred partner goals with goal-conditioned behavior models and planning (Huang et al., 2024).

The two objects are distinct but coupled. The current belief determines the partner model used in value evaluation, while the selected action affects which partner behaviors and outcomes are likely to be observed. The action therefore changes the distribution of future evidence, not the belief-update rule itself: the next belief is still obtained by applying τ_i to the evidence that becomes available. Functionally, the loop is

$$b_t^i \longrightarrow \underbrace{P_t^i = \text{Model}(b_t^i)}_{\text{partner model}} \longrightarrow a_t^i \longrightarrow e_{t+1}^i \longrightarrow b_{t+1}^i \longrightarrow P_{t+1}^i. \quad (6)$$

This is a functional decomposition for analysis, not an architectural requirement: the two objects need not be implemented as independent modules, and the equations alone do not assert a capability deficit in any particular agent.

3.2 BENAC-P: NEGOTIATION WITH PRIVATE PREFERENCES AND BINDING COMMITMENTS

We introduce BENAC-P, a multi-player negotiation game in which agents negotiate binding commitments toward shared public goals while holding private preferences over those goals. Building on BENAC’s commitment-based negotiation structure (Benac et al., 2026), we make partner incentives hidden and allow agents to propose menus of alternative agreements. A partner’s response both changes the feasible continuation and provides evidence about its preferences. This gives a concrete setting for the belief–planning loop above.

Game specification. A game is specified by

$$\mathcal{M} = \langle \mathcal{N}, \mathbf{K}, \mathcal{G}, \Gamma, \rho, C_0, \mathcal{L}, \mu_0 \rangle. \quad (7)$$

Here $\mathcal{N} = \{0, \dots, N-1\}$ is the player set, with $N \geq 3$ in our experiments, and $\mathbf{K} = (K_i)_{i \in \mathcal{N}}$ gives each player’s number of binary commitments. The set $\mathcal{G} = \{0, \dots, G-1\}$ indexes public goals; $\Gamma = (\Gamma_g)_{g \in \mathcal{G}}$ specifies the commitments required by each goal. The schedule $\rho = (\rho_0, \dots, \rho_{T-1})$ identifies the proposer at each turn, C_0 is the initial commitment state, and $\mathcal{L}$ specifies the legal proposals at each state. Finally, μ_0 is the publicly specified distribution over private preference configurations. All components of $\mathcal{M}$ are public; the realized preferences are not.

Commitments, preferences, and utility. At turn t , player i has commitments $c_{t,i} \in \{0, 1\}^{K_i}$, and $C_t = (c_{t,i})_{i \in \mathcal{N}}$ is the joint commitment state. An entry $c_{t,i,k} = 1$ records an irreversible commitment. Each requirement set $\Gamma_g \subseteq \{(i, k) : i \in \mathcal{N}, 1 \leq k \leq K_i\}$ involves at least two players. A goal is satisfied exactly when all its requirements hold:

$$S_g(C_t) = \mathbf{1}[c_{t,i,k} = 1 \text{ for every } (i, k) \in \Gamma_g]. \quad (8)$$

At initialization, the environment selects a preference matrix $V = (v_{i,g}) \sim \mu_0$, where $v_{i,g} \in \{-1, 0, +1\}$ represents AVOID, NEUTRAL, and WANT, respectively. Player i observes its own row v_i ; other rows remain private, although the public configuration distribution can correlate them. The episode ends after the scheduled interactions, and player i receives

$$U_i(C_T; v_i) = \sum_{g \in \mathcal{G}} v_{i,g} S_g(C_T). \quad (9)$$

There is no separately shaped intermediate reward. Agents therefore share the goal structure without necessarily sharing objectives: completing the same goal can benefit one player and harm another.

Offers, responses, and transitions. At turn t , proposer $p = \rho_t$ chooses a proposal from $\mathcal{L}(s_t)$, where $s_t = (C_t, t, h_t)$ contains the commitments, turn index, and public transcript. An ordinary offer to partner $q \neq p$ is $o = (q, \bar{c}_p, \bar{c}_q)$, specifying the two players’ target commitment vectors. Targets must preserve all existing commitments, add at most M new commitments per side, add no action in a public forbidden set F , and add at least one commitment in total. These constraints, together with any public stage-specific proposal restrictions, define $\mathcal{L}$.

The responder chooses ACCEPT or REJECT. Acceptance makes the proposed commitments immediately binding:

$$c_{t+1,i} = \begin{cases} c_{t,i} \vee \bar{c}_i, & i \in \{p, q\}, \\ c_{t,i}, & \text{otherwise.} \end{cases} \quad (10)$$

Rejection leaves C_t unchanged. A menu $m = (o^{(1)}, o^{(2)})$ contains two distinct legal offers to the same responder, who chooses CHOOSE_1, CHOOSE_2, or REJECT. Only the selected offer binds; rejection binds neither. Where allowed, PASS skips the proposal and leaves commitments unchanged. Each resolved interaction advances the schedule by one turn, and the proposal, response, and resulting state are publicly observed. A pending offer is hypothetical until accepted.

Although each agreement is bilateral, its commitments can enable or constrain later agreements with other players. A menu also lets ego choose which preference comparisons a partner faces. Information is consequently obtained through ordinary negotiation actions that have real task consequences, rather than through a separate preference-query action.

Deterministic rational partners. To diagnose ego behavior under a known response model, we use deterministic partners in the controlled subgames of Section 4. For a proposal a addressed to j , let $\mathcal{R}(a)$ be its legal response set and let $\mathsf{T}(s, a, r)$ be the public transition defined above. The partner compares responses using only public commitments and its own preferences:

$$f_j(s, a; v_j) \in \arg \max_{r \in \mathcal{R}(a)} \sum_{g \in \mathcal{G}} v_{j,g} S_g(C(\mathsf{T}(s, a, r))), \quad (11)$$

where $C(s)$ extracts the commitment state. Rejection wins ties with an agreement; among equally valuable accepted menu options, the earlier option wins. We certify each subgame by enumerating allowed continuations and verifying that this commitment-state ranking also maximizes the responder’s terminal utility. Thus, the snapshot comparison is terminal-rational in the certified family; it is not assumed to be sufficient in arbitrary negotiation games. The acting partner never reads another player’s private preference row. Uncertainty comes from hidden preferences, not randomized partner behavior.

Exact belief update and reference planning. For ego i , a hypothesis $h = (v_j)_{j \neq i}$ is a joint configuration of partner preferences. The solver initializes a distribution b_0^i by conditioning μ_0 on ego’s own row and any supplied initial information. Its support $\mathcal{H}_t^i$ retains cross-partner correlations. After ego proposes a and observes response r from j , the exact update is

$$b_{t+1}^i(h) = \frac{b_t^i(h) \mathbf{1}[f_j(s_t, a; v_j(h)) = r]}{\sum_{h' \in \mathcal{H}_t^i} b_t^i(h') \mathbf{1}[f_j(s_t, a; v_j(h')) = r]}. \quad (12)$$

Inconsistent configurations are eliminated; ego’s proposal itself is an intervention, not evidence about hidden preferences. Our diagnostic population is uniform over its compatible configurations, so this update reduces to exact support filtering. The model-facing belief can therefore be expressed as a semantic set without requiring numerical probabilities.

The reference planner enumerates the remaining interaction tree. In our diagnostic subgames, ego makes the scheduled proposals and partners supply the responses, including the final approval. Define $p_b(r \mid s, a) = \sum_h b(h) \mathbf{1}[f_j(s, a; v_j(h)) = r]$ and let $\tau(b, s, a, r)$ denote the update above. The solver computes

$$W_i(s, b) = \max_{a \in \mathcal{L}(s)} \sum_{r \in \mathcal{R}(a)} p_b(r \mid s, a) W_i(\mathsf{T}(s, a, r), \tau(b, s, a, r)), \quad W_i(s_T, b) = U_i(C_T; v_i), \quad (13)$$

with zero-probability branches omitted. A pass has one null response and leaves belief unchanged; an automatic protocol step has a singleton action set. This solver selects actions by terminal utility after observation, updating, and replanning. It provides exact judgments and reference actions for diagnosis under the specified partner policies, without giving ego access to the realized hidden configuration or requiring an equilibrium solution.

4 DIAGNOSE

We use a compact controlled diagnosis to identify targets for post-training: *partner belief* (B), the formation and updating of task-relevant partner judgments, and *partner-conditioned planning* (P), the selection of interaction actions using those judgments. We separately test each capability and intervene on their interfaces to examine the loop $B_t \rightarrow P_t \rightarrow e_{t+1} \rightarrow B_{t+1}$. The evidence-generating consequence of an action is part of planning, rather than a third capability.

4.1 CONTROLLED INTERACTION AND DIAGNOSTIC INTERVENTIONS

Task construction. We instantiate a restricted three-stage BENAC-P protocol with three players: ego P_0 , an assessment-and-permit partner P_1 , and an approval partner P_2 . Ego first chooses a preliminary interaction with P_1 , including preparation, an assessment offer, a two-offer menu, or skipping. It then negotiates a binding route and permit with P_1 ; finally, the bound route is automatically submitted to P_2 for approval. Project goals require all three players’ commitments. Assessments incur an ego cost, preparation yields a benefit, and some preliminary menus already lock a route. Thus,

information can affect a later decision, but acquiring it can also consume resources or irreversibly restrict that decision.

The public specification gives three joint preference configurations. P_1 's preference for the first assessment is respectively WANT, NEUTRAL, or AVOID, with opposite preferences for the second assessment in the two non-neutral configurations. In the main condition, these configurations also determine which route P_2 will approve. A menu offering the two assessments can therefore distinguish the configurations through its chosen option or rejection. Partners respond deterministically, maximize their own terminal utility, and access only public state and their own preferences. We certify response optimality over all allowed continuations. Hidden configurations remain unknown to ego; there is no random partner error.

Each instance has three matched conditions: *unknown/relevant*, where partner information affects the best route; *known*, where the initial configuration is supplied; and *unknown/irrelevant*, where P_2 approves every route. Enumeration certifies that the informative assessment menu is optimal in the first condition, whereas preparation strictly dominates it in the two controls. The reference planner maximizes expected *terminal task utility*, including subsequent observation and replanning, rather than information gain alone. These are selected instances for conditional diagnosis, not a sample estimating failure prevalence in unrestricted BENAC games.

Semantic interface and four tests. The model submits either a nonempty subset of $\{\text{WANT, NEUTRAL, AVOID}\}$ for P_1 's first assessment, or the index of a listed legal action. The subset identifies compatible joint configurations through the public table; the backend uses their stated equal frequencies. Models do not output probabilities, utilities, Q values, or plan variables. The prompt distinguishes the population configuration table from the episode-specific initial support.

B alone: We provide the initial support and a legal history and ask for the still-possible preference set. The reference set contains exactly the configurations compatible with the deterministic responses. We measure exact set agreement and the fraction of ruled-out labels incorrectly retained. Initial judgments and post-response judgments are reported separately.

P given correct B: We supply the reference judgment and ask for an action, both at the initial interaction and after a partner response. Regret is the difference from the reference-optimal expected terminal utility, using reference continuation after the chosen action; all optimal ties receive zero regret. This isolates action selection conditional on a correct supplied judgment from the model's own belief-estimation errors.

B→P: At the same decision state, we replace the model's judgment with the reference judgment while holding the planner, legal actions, and physical state fixed. Both planning queries use fresh contexts with the same judgment format, without the original history or belief-generation scratch-pad. The state description retains decision-relevant commitments but omits historical details that would independently reveal the response. Repair supplies only what the evidence supports, not the hidden configuration. The paired reduction in regret measures whether better partner understanding improves this model's action selection.

P→B: From the same initial state and correct initial judgment, we force either the model-selected or reference-selected first action. We enumerate every possible response, obtain the same model's updated judgment, and evaluate downstream utility with a reference planner. We separately measure the action's evidence channel through expected reference-posterior entropy and the model's use of that evidence through judgment and updater-repair scores. Response branches are weighted by compatible configurations, not selected for successful outcomes. Because actions also alter commitments, a utility difference alone does not identify an information-mediated effect.

4.2 RESULTS AND IMPLICATIONS FOR TRAINING

We evaluate Qwen3-4B-Instruct-2507 at temperature zero on 12 instance bundles, each containing the three conditions. We report the six bundles designated *confirmation*; the remaining six are designated *discovery*. These are within-family diagnostic results, not an untouched generalization test: the instance split was reused during interface development. Intervals are 95% percentile bootstrap intervals over the six bundles (2,000 resamples), after weighting response branches within each game. All 486 diagnostic tasks produced valid submissions; 31 initially reached the 1,024-

Table 1: Diagnostic results in the unknown/relevant condition, six confirmation bundles. Utility quantities use additive goal-utility units. B accuracy and belief repair concern response branches of the reference first action; initial interaction regret is reported separately.

Measure	Estimate	95% interval
Post-response B exact ($\uparrow$)	22.2%	[11.1, 33.3]%
Ruled-out labels retained ($\downarrow$)	77.8%	[66.7, 88.9]%
Initial P regret given correct B ($\downarrow$)	2.556	[1.333, 3.778]
Post-response P regret given correct B ($\downarrow$)	0.667	[0.000, 1.444]
B repair: reduction in model-planner regret	2.167	[1.219, 3.111]
Action repair: additional channel information (bits)	1.057	[0.528, 1.585]
Updater repair after reference action: utility gain	2.889	[2.167, 3.722]
Action repair with model updater: utility gain	0.444	[-1.333, 2.500]

token limit and were completed through one additional submission request capped at 128 tokens. All initial responses contained reasoning, with a mean of 301.5 total completion tokens per task including recovery. Four separate empty-history interface checks passed before the diagnostic run. Appendix A provides protocol details and the factorial tables.

Distinct weaknesses with task consequences. Initial B judgments are exact in all three confirmation conditions, resolving the basic concern that the model cannot preserve the supplied initial support. However, after the informative reference action, exact agreement falls to 22.2% (Table 1). The model retains 77.8% of labels that responses rule out, without excluding any truly supported label in this condition. This is predominantly insufficient elimination of alternatives. For example, after the first assessment is chosen from a menu, one response acknowledges the choice but treats all three configurations as still possible, failing to apply the public preference and tie-breaking rules. Planning also fails with correct B: initial interaction regret is 2.556. The smaller post-response regret has an interval including zero; the stronger planning evidence concerns choosing the interaction itself. Replacing model B with reference B reduces model-planner regret by 2.167, establishing a beneficial B $\rightarrow$ P intervention in these instances.

Better evidence is not automatically used. Replacing the first action increases available information by 1.057 bits. Nevertheless, reference-action and model-action branches both have mean model B exact agreement of 22.2%, despite different reference uncertainty. After the reference action, replacing the model updater with the reference updater improves downstream reference-planner utility by 2.889. Action repair alone, retaining the model updater, has an uncertain utility benefit of 0.444. Thus, the intervention establishes control of the evidence channel, while the updater remains a bottleneck to realizing its value; it does not establish a reliable end-to-end benefit from action repair alone.

The controls clarify the scope. In the known condition, post-response B exact is 50.0% despite perfect initial judgments, revealing failures to preserve known information across an otherwise uninformative interaction. In the unknown/irrelevant condition, post-response planning regret and B-repair value are both zero: downstream route value is independent of partner configuration by construction. Together, these measurements motivate targeted training of both partner understanding and conditioned planning, followed by evaluating their composition and transfer. They do not establish a universal deficit across models or environments.

5 TRAINING

6 TEST AND TRANSFER

7 DISCUSSION

8 CONCLUSION

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ACKNOWLEDGMENTS

Use unnumbered third level headings for the acknowledgments. All acknowledgments, including those to funding agencies, go at the end of the paper.

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A DIAGNOSTIC PROTOCOL AND DECOMPOSITION

Execution and aggregation. The run uses seeds 20000–20011; 20000–20005 are the designated discovery bundles and 20006–20011 the designated confirmation bundles. Each bundle contains unknown/relevant, known, and unknown/irrelevant games, with shared underlying instance parameters. Action ordering and route names vary with seed. Responses are exhaustively enumerated under the publicly specified finite population, so partner-behavior sampling is not a source of variance. The uncertainty intervals quantify variation among six bundles within this selected family; neither response branches nor model requests are treated as independent games. Reuse of this split during prompt/interface development precludes interpreting the intervals as an untouched confirmatory test.

The model is instructed to briefly reason, with soft targets of 120 words for judgment questions and 240 for action questions. The initial request exposes one native submission tool with automatic tool choice. A length-stopped response receives exactly one additional request containing the original question and its generated text, followed by an instruction to submit through the named function without further analysis. No reference answer or corrective feedback is supplied. Initial truncation remains recorded even after successful submission. There are 517 actual requests for 486 diagnostic tasks, consuming 146,553 completion tokens; the four preliminary checks consume another 579 completion tokens in four requests and are excluded from the diagnostic statistics. Those checks use empty histories with singleton WANT, NEUTRAL, AVOID, and the unresolved three-label support. They test interface semantics, not interaction competence. Invalid outputs would not be converted to actions; no such outputs remained in this run.

Regret decomposition. Let O denote the reference component and L the evaluated model. For a post-response state, R_{XY} uses judgment provider X and planner Y ; regret is evaluated under the evidence-supported reference judgment, even when the planner receives model judgment. The reference planner optimizes under the judgment actually supplied to it. Thus $R_{OO} = 0$, R_{OL} measures planning deficit with correct judgment, R_{LO} measures judgment error’s task cost under reference planning, and $R_{LL} - R_{OL}$ is the model-planner benefit of judgment repair. We average over all responses to the reference first action.

For the reverse intervention, J_{XY} uses initial action chooser X and subsequent judgment updater Y , followed by reference planning, and reports expected terminal utility. The initial judgment is correct in both action arms. $J_{OL} - J_{LL}$ is the effect of action repair with the model updater; $J_{OO} - J_{OL}$ and $J_{LO} - J_{LL}$ are updater-repair effects on the reference and model actions, respectively. With latent configuration Z and response Y_a , channel information is $H(Z) - \mathbb{E}[H(Z \mid Y_a, a)]$ in bits. This is a backend measurement, not an action-selection reward.

Table 2: Two component decompositions in confirmation unknown/relevant games. The left table reports regret (lower is better); the right reports terminal utility (higher is better). Means over six bundles.

<i>R</i> : judgment / planner	Reference	Model	<i>J</i> : chooser / updater	Reference	Model
Reference	0.000	0.667	Reference	4.833	1.944
Model	2.889	2.833	Model	2.278	1.500

Table 3: Matched confirmation controls. B exact and downstream regret use reference-action response branches. Initial regret uses correct initial B. Downstream indifference in the irrelevant condition is a designed control.

Condition	Post-response B exact	Initial P regret	Downstream P regret
Unknown/relevant	22.2%	2.556	0.667
Known	50.0%	2.333	0.833
Unknown/irrelevant	33.3%	0.389	0.000

Certification and interpretation. The allowed-action catalogue is public and stage-specific; the reference optimizer does not search unrestricted BENAC offers. Certification checks that every prescribed partner response maximizes its own terminal utility for every allowed continuation, that the informative menu is optimal in the unknown/relevant condition, and that preparation strictly dominates that menu in both controls. A further same-menu comparison verifies positive task value from updating the reference judgment rather than freezing the initial one. This establishes an information-use opportunity without treating all between-action utility differences as information effects. Equal-utility reference actions use a fixed enumeration tie-break; all ties remain correct for model action scoring. The modular end-to-end regret is 3.611 [1.611, 5.611] in unknown/relevant confirmation games; it is a task-level outcome rather than a separate capability diagnostic.

B APPENDIX

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The style files for ICLR and other conference information are available online at:

<http://www.iclr.cc/>

The file `iclr2027_conference.pdf` contains these instructions and illustrates the various formatting requirements your ICLR paper must satisfy. Submissions must be made using L^AT_EX and the style files `iclr2027_conference.sty` and `iclr2027_conference.bst` (to be used with L^AT_EX2e). The file `iclr2027_conference.tex` may be used as a “shell” for writing your paper. All you have to do is replace the author, title, abstract, and text of the paper with your own.

The formatting instructions contained in these style files are summarized in sections D, E, and F below.

D GENERAL FORMATTING INSTRUCTIONS

The text must be confined within a rectangle 5.5 inches (33 picas) wide and 9 inches (54 picas) long. The left margin is 1.5 inch (9 picas). Use 10 point type with a vertical spacing of 11 points. Times New Roman is the preferred typeface throughout. Paragraphs are separated by 1/2 line space, with no indentation.

Paper title is 17 point, in small caps and left-aligned. All pages should start at 1 inch (6 picas) from the top of the page.

Authors' names are set in boldface, and each name is placed above its corresponding address. The lead author's name is to be listed first, and the co-authors' names are set to follow. Authors sharing the same address can be on the same line.

Please pay special attention to the instructions in section F regarding figures, tables, acknowledgments, and references.

There will be a strict upper limit of **9 pages** for the main text of the initial submission, with unlimited additional pages for citations. This limit will be expanded to **10 pages** for rebuttal/camera ready.

E HEADINGS: FIRST LEVEL

First level headings are in small caps, flush left and in point size 12. One line space before the first level heading and 1/2 line space after the first level heading.

E.1 HEADINGS: SECOND LEVEL

Second level headings are in small caps, flush left and in point size 10. One line space before the second level heading and 1/2 line space after the second level heading.

E.1.1 HEADINGS: THIRD LEVEL

Third level headings are in small caps, flush left and in point size 10. One line space before the third level heading and 1/2 line space after the third level heading.

F CITATIONS, FIGURES, TABLES, REFERENCES

These instructions apply to everyone, regardless of the formatter being used.

F.1 CITATIONS WITHIN THE TEXT

Citations within the text should be based on the `natbib` package and include the authors' last names and year (with the "et al." construct for more than two authors). When the authors or the publication are included in the sentence, the citation should not be in parenthesis using `\citet{}` (as in "See ? for more information."). Otherwise, the citation should be in parenthesis using `\citep{}` (as in "Deep learning shows promise to make progress towards AI (?).").

The corresponding references are to be listed in alphabetical order of authors, in the REFERENCES section. As to the format of the references themselves, any style is acceptable as long as it is used consistently.

Table 4: Sample table title

PART	DESCRIPTION
Dendrite	Input terminal
Axon	Output terminal
Soma	Cell body (contains cell nucleus)

F.2 FOOTNOTES

Indicate footnotes with a number¹ in the text. Place the footnotes at the bottom of the page on which they appear. Precede the footnote with a horizontal rule of 2 inches (12 picas).²

F.3 FIGURES

All artwork must be neat, clean, and legible. Lines should be dark enough for purposes of reproduction; art work should not be hand-drawn. The figure number and caption always appear after the figure. Place one line space before the figure caption, and one line space after the figure. The figure caption is lower case (except for first word and proper nouns); figures are numbered consecutively.

Make sure the figure caption does not get separated from the figure. Leave sufficient space to avoid splitting the figure and figure caption.

You may use color figures. However, it is best for the figure captions and the paper body to make sense if the paper is printed either in black/white or in color.

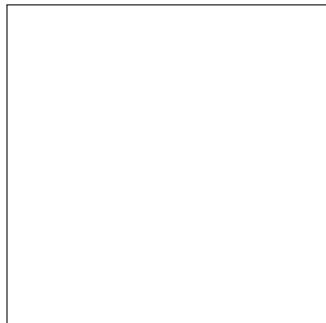


Figure 1: Sample figure caption.

F.4 TABLES

All tables must be centered, neat, clean and legible. Do not use hand-drawn tables. The table number and title always appear before the table. See Table 4.

Place one line space before the table title, one line space after the table title, and one line space after the table. The table title must be lower case (except for first word and proper nouns); tables are numbered consecutively.

G DEFAULT NOTATION

In an attempt to encourage standardized notation, we have included the notation file from the textbook, *Deep Learning* ? available at https://github.com/goodfeli/dlbook_

¹Sample of the first footnote

²Sample of the second footnote

notation/. Use of this style is not required and can be disabled by commenting out `math_commands.tex`.

Numbers and Arrays

a	A scalar (integer or real)
$\mathbf{a}$	A vector
$\mathbf{A}$	A matrix
$\mathbf{A}$	A tensor
$\mathbf{I}_n$	Identity matrix with n rows and n columns
$\mathbf{I}$	Identity matrix with dimensionality implied by context
$\mathbf{e}^{(i)}$	Standard basis vector $[0, \dots, 0, 1, 0, \dots, 0]$ with a 1 at position i
$\text{diag}(\mathbf{a})$	A square, diagonal matrix with diagonal entries given by $\mathbf{a}$
$\mathbf{a}$	A scalar random variable
$\mathbf{a}$	A vector-valued random variable
$\mathbf{A}$	A matrix-valued random variable

Sets and Graphs

$\mathbb{A}$	A set
$\mathbb{R}$	The set of real numbers
$\{0, 1\}$	The set containing 0 and 1
$\{0, 1, \dots, n\}$	The set of all integers between 0 and n
$[a, b]$	The real interval including a and b
$(a, b]$	The real interval excluding a but including b
$\mathbb{A} \setminus \mathbb{B}$	Set subtraction, i.e., the set containing the elements of $\mathbb{A}$ that are not in $\mathbb{B}$
$\mathcal{G}$	A graph
$\text{Pa}_{\mathcal{G}}(\mathbf{x}_i)$	The parents of $\mathbf{x}_i$ in $\mathcal{G}$

Indexing

a_i	Element i of vector $\mathbf{a}$, with indexing starting at 1
$\mathbf{a}_{-i}$	All elements of vector $\mathbf{a}$ except for element i
$\mathbf{A}_{i,j}$	Element i, j of matrix $\mathbf{A}$
$\mathbf{A}_{i,:}$	Row i of matrix $\mathbf{A}$
$\mathbf{A}_{:,i}$	Column i of matrix $\mathbf{A}$
$\mathbf{A}_{i,j,k}$	Element (i, j, k) of a 3-D tensor $\mathbf{A}$
$\mathbf{A}_{:,:,i}$	2-D slice of a 3-D tensor
$\mathbf{a}_i$	Element i of the random vector $\mathbf{a}$

Calculus

648	$\frac{dy}{dx}$	Derivative of y with respect to x
649	$\frac{\partial y}{\partial x}$	Partial derivative of y with respect to x
650	$\nabla_x y$	Gradient of y with respect to x
651	$\nabla_{\mathbf{X}} y$	Matrix derivatives of y with respect to $\mathbf{X}$
652	$\nabla_{\mathbf{x}} y$	Tensor containing derivatives of y with respect to $\mathbf{X}$
653	$\frac{\partial f}{\partial \mathbf{x}}$	Jacobian matrix $\mathbf{J} \in \mathbb{R}^{m \times n}$ of $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$
654	$\nabla_{\mathbf{x}}^2 f(\mathbf{x})$ or $\mathbf{H}(f)(\mathbf{x})$	The Hessian matrix of f at input point $\mathbf{x}$
655	$\int f(\mathbf{x}) d\mathbf{x}$	Definite integral over the entire domain of $\mathbf{x}$
656	$\int_{\mathbb{S}} f(\mathbf{x}) d\mathbf{x}$	Definite integral with respect to $\mathbf{x}$ over the set $\mathbb{S}$
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658		Probability and Information Theory
659	$P(\mathbf{a})$	A probability distribution over a discrete variable
660	$p(\mathbf{a})$	A probability distribution over a continuous variable, or over a variable whose type has not been specified
661	$\mathbf{a} \sim P$	Random variable $\mathbf{a}$ has distribution P
662	$\mathbb{E}_{\mathbf{x} \sim P}[f(\mathbf{x})]$ or $\mathbb{E}f(\mathbf{x})$	Expectation of $f(\mathbf{x})$ with respect to $P(\mathbf{x})$
663	$\text{Var}(f(\mathbf{x}))$	Variance of $f(\mathbf{x})$ under $P(\mathbf{x})$
664	$\text{Cov}(f(\mathbf{x}), g(\mathbf{x}))$	Covariance of $f(\mathbf{x})$ and $g(\mathbf{x})$ under $P(\mathbf{x})$
665	$H(\mathbf{x})$	Shannon entropy of the random variable $\mathbf{x}$
666	$D_{\text{KL}}(P \ Q)$	Kullback-Leibler divergence of P and Q
667	$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$	Gaussian distribution over $\mathbf{x}$ with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$
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669		Functions
670	$f : \mathbb{A} \rightarrow \mathbb{B}$	The function f with domain $\mathbb{A}$ and range $\mathbb{B}$
671	$f \circ g$	Composition of the functions f and g
672	$f(\mathbf{x}; \boldsymbol{\theta})$	A function of $\mathbf{x}$ parametrized by $\boldsymbol{\theta}$. (Sometimes we write $f(\mathbf{x})$ and omit the argument $\boldsymbol{\theta}$ to lighten notation)
673	$\log x$	Natural logarithm of x
674	$\sigma(x)$	Logistic sigmoid, $\frac{1}{1 + \exp(-x)}$
675	$\zeta(x)$	Softplus, $\log(1 + \exp(x))$
676	$\ \mathbf{x}\ _p$	L^p norm of $\mathbf{x}$
677	$\ \mathbf{x}\ $	L^2 norm of $\mathbf{x}$
678	x^+	Positive part of x , i.e., $\max(0, x)$
679	$\mathbf{1}_{\text{condition}}$	is 1 if the condition is true, 0 otherwise
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H FINAL INSTRUCTIONS

Do not change any aspects of the formatting parameters in the style files. In particular, do not modify the width or length of the rectangle the text should fit into, and do not change font sizes (except perhaps in the REFERENCES section; see below). Please note that pages should be numbered.

I PREPARING POSTSCRIPT OR PDF FILES

Please prepare PostScript or PDF files with paper size “US Letter”, and not, for example, “A4”. The `-t letter` option on `dvips` will produce US Letter files.

Consider directly generating PDF files using `pdflatex` (especially if you are a MiKTeX user). PDF figures must be substituted for EPS figures, however.

Otherwise, please generate your PostScript and PDF files with the following commands:

```
dvips mypaper.dvi -t letter -Ppdf -G0 -o mypaper.ps
ps2pdf mypaper.ps mypaper.pdf
```

I.1 MARGINS IN LATEX

Most of the margin problems come from figures positioned by hand using `\special` or other commands. We suggest using the command `\includegraphics` from the `graphicx` package. Always specify the figure width as a multiple of the line width as in the example below using `.eps` graphics

```
\usepackage[dvips]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.eps}
```

or

```
\usepackage[pdftex]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.pdf}
```

for `.pdf` graphics. See section 4.4 in the graphics bundle documentation (<http://www.ctan.org/tex-archive/macros/latex/required/graphics/grfguide.ps>)

A number of width problems arise when LaTeX cannot properly hyphenate a line. Please give LaTeX hyphenation hints using the `\-` command.